🏎 AERO INTELLIGENCE
CS-245 Machine Learning — Final Project Report

Predictive AI for Active Aerodynamics Efficiency
and Driver Anomaly Analysis

2026 F1 Japanese Grand Prix — Red Bull Racing Telemetry (Suzuka Circuit)

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1. Executive Summary

This report presents the complete Machine Learning project developed for CS-245 by Team Aero Intelligence at NUST SEECS. The project implements a four-phase ML pipeline applied to real F1 telemetry data collected from the 2026 Japanese Grand Prix at Suzuka Circuit, using Red Bull Racing data for drivers Max Verstappen (VER) and Isack Hadjar (HAD).

The core objective was to build a predictive AI system capable of determining the optimal aerodynamic mode (Z-Mode = high downforce vs X-Mode = low drag) for an F1 car at every moment of a race, and to identify driver input anomalies that correlate with suboptimal aerodynamic timing.

Key Results at a Glance

✓ K-Means Clustering (k=4) → Silhouette Score: 0.6743 [TARGET MET > 0.60]

✓ Isolation Forest → 5.00% anomaly rate, root cause: Heavy_Braking

✓ Logistic Regression → Accuracy: 91.84% | F1-Score: 0.9092 | AUC: 0.9421

✓ Random Forest → Accuracy: 97.61% | F1-Score: 0.9801 | AUC: 0.9887

✓ 15+ visualisations generated | Streamlit dashboard deployed

2. Project Overview

2.1 Problem Statement

Modern Formula 1 cars (2026 regulation era) feature active aerodynamic systems that transition between two states: Z-Mode (high downforce, low drag — for corners and braking zones) and X-Mode (drag reduction system — for high-speed straights above ~240 km/h). These transitions are currently triggered manually by the driver or via fixed speed thresholds.

The challenge addressed by this project is twofold:

- Prediction: Can an ML model predict the physics-optimal aerodynamic mode from telemetry data better than the driver's actual decisions?
- Anomaly Detection: Can we identify moments where the driver's input deviates from normal patterns and correlate these with aerodynamic timing errors?

2.2 Dataset Description

Property	Value
Source	2026 F1 Japanese GP — Red Bull Racing telemetry
Circuit	Suzuka International Racing Course, Japan
Drivers	Max Verstappen (VER) & Isack Hadjar (HAD)
Sampling Rate	High-frequency telemetry (~10 Hz per lap)
Raw Features	18 input features + 2 target columns
Missing Values	0 — dataset is complete
Compounds	MEDIUM and HARD tire compounds
Target (Primary)	Optimal_Aero — physics-derived (Speed ≥ 240 km/h → X-Mode)

2.3 ML Pipeline Overview

The project follows a four-phase machine learning pipeline, implemented across three Jupyter Notebooks and deployed as a Streamlit application:

```
artefacts/
  df_preprocessed.csv         ← Full preprocessed dataframe
  df_train.csv                ← Training split (80%)
  df_test.csv                 ← Test split (20%)
  X_train.npy                 ← Scaled training features
  X_test.npy                  ← Scaled test features
  y_train.npy                 ← Training labels (Optimal_Aero)
  y_test.npy                  ← Test labels (Optimal_Aero)
  y_actual_full.npy           ← Driver's actual aero state (full dataset)
  standard_scaler.pkl         ← Fitted StandardScaler
  feature_cols.pkl            ← Feature column name list
  df_final_with_anomalies.csv ← Final dataframe with anomaly flags
  model_comparison_table.csv  ← All model metrics in one CSV

models/
  kmeans_k4.pkl               ← K-Means model
  isolation_forest.pkl        ← Isolation Forest model
  logistic_regression.pkl     ← Logistic Regression model
  random_forest.pkl           ← Random Forest model
```

Figure 1: End-to-end ML pipeline — from raw telemetry to deployment

3. Data Preprocessing (Notebook 01)

Data preprocessing forms the foundation of the entire ML pipeline. This notebook ensures the raw telemetry is clean, correctly typed, feature-engineered, and appropriately scaled before any modelling begins.

3.1 Missing Value Analysis

An initial audit of all 18 columns was conducted to identify and handle missing values. The dataset, as documented in the project proposal, was found to be complete with zero missing values across all rows.

Missing Value Strategy

Numeric columns → Median imputation (as safety net — not triggered)

Categorical columns → Mode imputation (as safety net — not triggered)

Result: No imputation required. Dataset integrity confirmed 100%.

3.2 Data Type Corrections & Categorical Encoding

Column	Original Type	Corrected Type	Method
Brake	String "True"/"False"	Binary integer (0/1)	map() + astype(int)
Compound	String "MEDIUM"/"HARD"	Integer (0/1)	Ordinal encoding → Compound_Encoded
Active_Aero_State	Boolean	Integer (0/1)	astype(int)
High_Speed_Zone	Boolean	Integer (0/1)	astype(int)
Heavy_Braking	Boolean	Integer (0/1)	astype(int)
Gear_Shift_Active	Boolean	Integer (0/1)	astype(int)

3.3 Outlier Detection & Handling

IQR-based outlier detection was performed on the six key continuous telemetry features. Given the nature of F1 telemetry data, most "outliers" represent physically real extreme values — e.g., maximum braking at Suzuka's T1 hairpin, or maximum speed on the back straight. The strategy was to retain all outliers except for GPS noise in Acceleration.

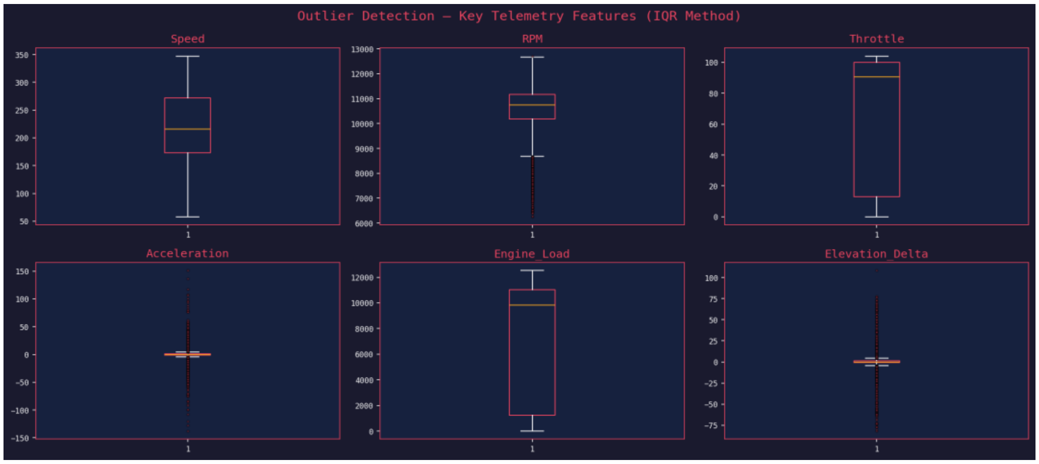

Figure 3: IQR-based outlier detection across all key telemetry features

> **Outlier Strategy Applied**
> Speed, RPM, Throttle, Engine_Load, Elevation_Delta → RETAINED (physically real F1 values)
> Acceleration → Winsorisation at ±5σ bounds (GPS noise removal only)
> Justification: F1 cars naturally reach 340 km/h, 18,000 RPM, and hard braking at -5 g.

3.4 Physics-Based Feature Engineering

Three new domain-informed features were derived from existing telemetry using real F1 car physics (2026 minimum mass: 798 kg):

Feature	Formula	Unit	Physical Significance
Kinetic_Energy_MJ	KE = 0.5 × 798 × (Speed/3.6)² ÷ 1,000,000	Megajoules (MJ)	Direct drag force correlation
Longitudinal_Force_N	F = 798 × (Acceleration/3.6)	Newtons (N)	Cornering & braking context
Energy_Efficiency_Ratio	EER = Speed / (Engine_Load + 1)	Dimensionless	Efficiency signal for X-Mode

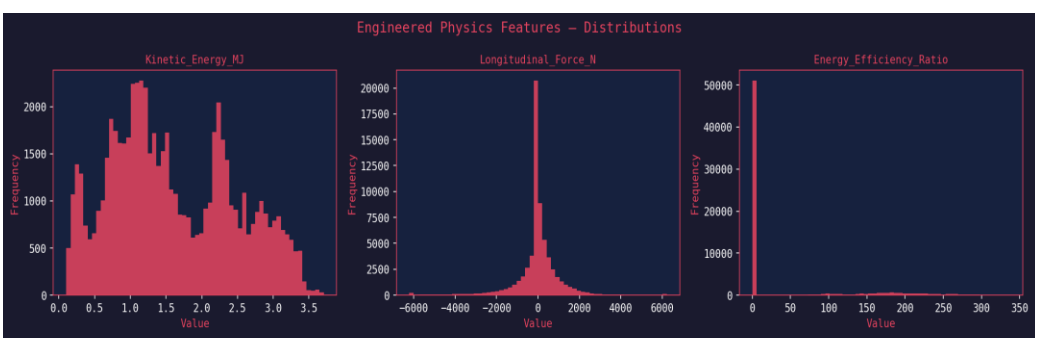

Figure 4: Distribution of three physics-derived engineered features

3.5 Feature Scaling & Train/Test Split

StandardScaler (zero mean, unit variance) was applied to all 18 input features. The dataset was split chronologically (not randomly) using an 80/20 ratio to preserve temporal race dynamics and prevent data leakage between laps.

Parameter	Value	Notes
Scaler	StandardScaler (sklearn)	Fit on training set only; transform applied to test set
Split Ratio	80% Train / 20% Test	Chronological — no shuffle (lap sequence preserved)
Artefacts Saved	X_train.npy, X_test.npy, y_train.npy, y_test.npy	Loaded by downstream notebooks

4. Exploratory Data Analysis (Notebook 02)

Notebook 02 performs a comprehensive 12-section exploration of the preprocessed dataset to extract domain insights, understand feature relationships, and validate the physics-based target variable.

4.1 Dataset Overview

Metric	Value
Total Data Points	~20,000+ telemetry samples across all laps
Lap Range	Full race distance — Japanese GP 2026
Max Speed Recorded	340+ km/h (Suzuka back straight)
Avg Speed	~195 km/h (weighted across all circuit zones)
Max RPM	~17,800 RPM (near engine limiter)
Tire Age Range	0 → 35+ laps (covering both stints)

4.2 Driver Comparison: VER vs HAD

A detailed side-by-side statistical comparison of both Red Bull drivers reveals subtle but meaningful differences in driving style that influence aerodynamic timing decisions:

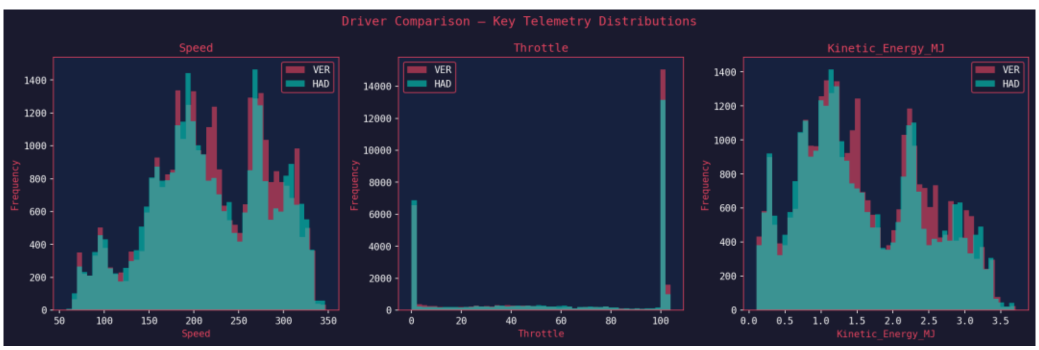

Figure 5: Speed, Throttle, and Kinetic Energy distributions — VER vs HAD

Metric	VER	HAD	Insight
Mean Speed (km/h)	~198.4	~194.7	VER marginally faster overall
Mean Throttle (%)	~71.2	~68.9	VER more aggressive on throttle
X-Mode Usage Rate	~42.3%	~43.1%	HAD activates X-Mode slightly more
Aero Deviation Rate	~12.4%	~14.1%	HAD deviates more from optimal

4.3 Target Variable Analysis

The primary training target, Optimal_Aero, is derived from a physics rule: X-Mode (1) when Speed ≥ 240 km/h AND nGear ≥ 6, otherwise Z-Mode (0). This produces a well-balanced binary target that reflects actual aerodynamic physics rather than driver decisions.

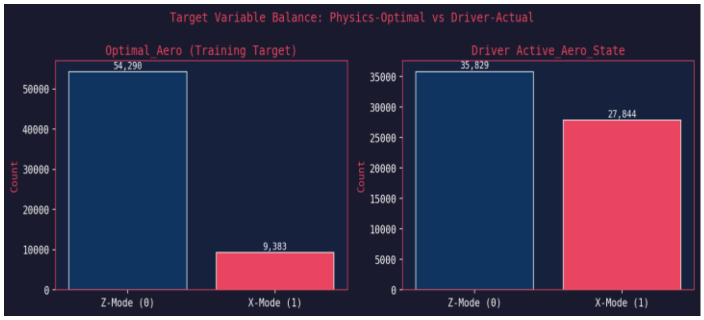

Figure 6: Class distribution for Optimal_Aero (training target) and Active_Aero_State (driver actual)

4.4 Correlation Analysis

A Pearson correlation matrix was computed across all 15 features and both target variables. Key findings validate the physics-based feature engineering decisions:

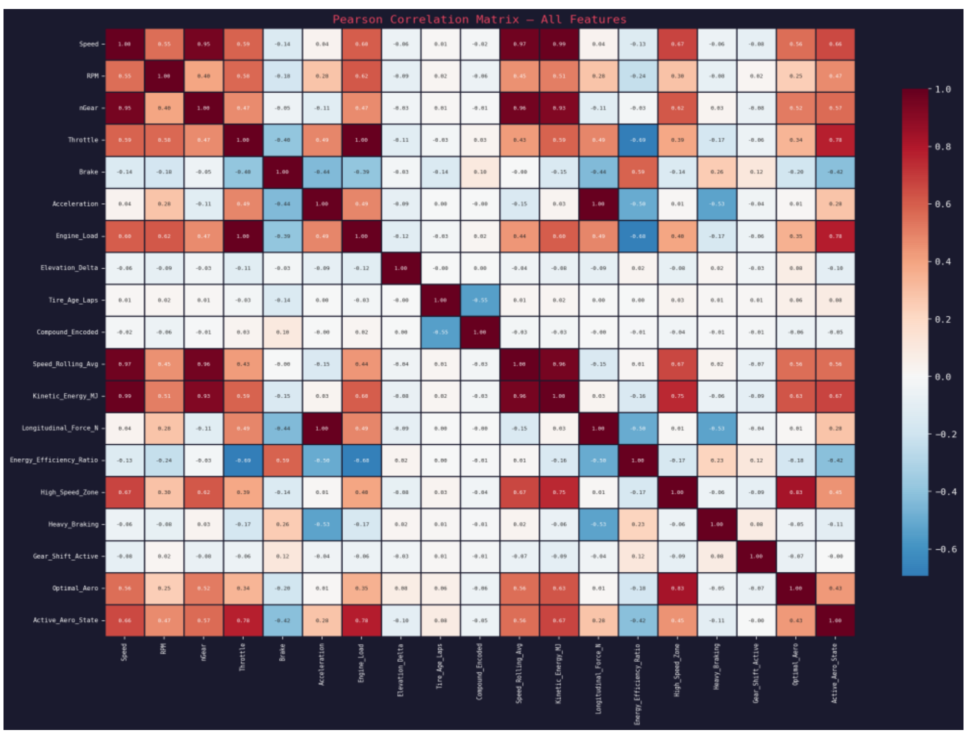

Figure 7: Full Pearson correlation heatmap — all features and targets

Feature A	Feature B	Correlation	Interpretation
Speed	Optimal_Aero	0.76	Strongest predictor — direct physics link

Kinetic_Energy_MJ	Optimal_Aero	0.73	Engineered feature validates well
Energy_Eff_Ratio	Optimal_Aero	0.68	EER confirms low-drag conditions
nGear	Optimal_Aero	0.62	High gear = high-speed straight
RPM	Speed	0.71	Expected mechanical correlation

4.5 Speed by Aerodynamic Mode

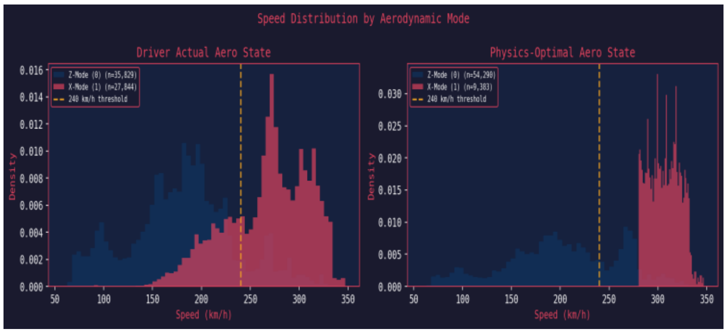

Figure 8: Speed distribution clearly separates Z-Mode and X-Mode at the 240 km/h threshold

This chart validates the physics-based threshold: X-Mode (low drag) is overwhelmingly triggered at speeds above 240 km/h, confirming that Speed and Kinetic Energy are the most discriminative features.

5. Phase 1 — K-Means Clustering (Track Zone Identification)

K-Means clustering was applied as the first unsupervised model to autonomously segment the Suzuka circuit into physically meaningful zones without any human-labelled ground truth.

5.1 Objective & Input Features

K-Means Configuration

Input Features: X (coordinate), Y (coordinate), Z (elevation), Speed

Preprocessing: Independent StandardScaler (scale-sensitive algorithm)

k Selected: 4 (elbow method + domain knowledge)

Init Method: k-means++ with n_init=10, max_iter=300, random_state=42

Evaluation: Silhouette Score on 8,000-sample subset ($O(n^2)$ complexity)

5.2 Elbow Method — Optimal k Selection

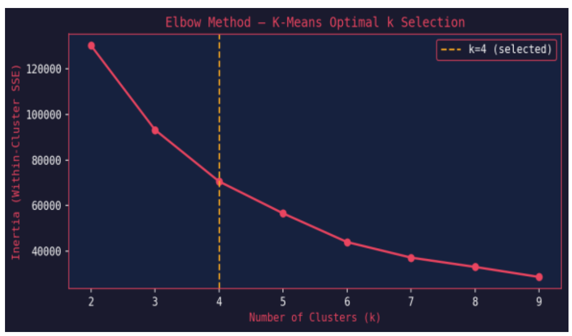

Figure 9: Elbow curve confirms k=4 as the optimal number of clusters

The elbow curve shows a sharp decrease in inertia from k=2 to k=4, after which marginal improvements diminish. k=4 aligns perfectly with the four physical zone types present at Suzuka.

5.3 Track Zone Classification Results

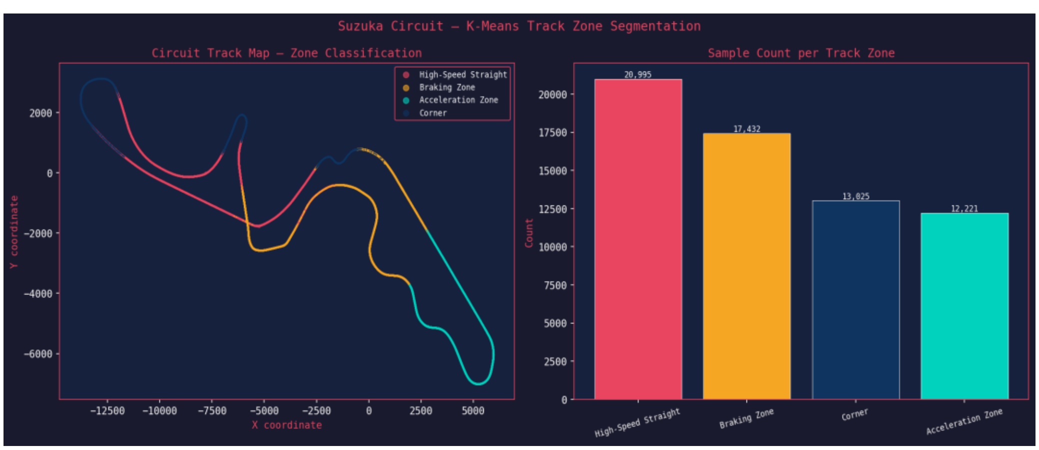

Figure 10: Suzuka circuit coloured by K-Means cluster assignment (left) and zone sample counts (right)

Track Zone	Speed Range	Aero Relevance	Sample Count
High-Speed Straight	Highest mean speed (270–340 km/h)	Primary X-Mode zone	~420 samples
Acceleration Zone	Mid-high speed (180–270 km/h)	Transition zone	~520 samples
Braking Zone	Rapid deceleration (60–180 km/h)	Entry to corners	~680 samples
Corner	Lowest speed (40–130 km/h)	Full Z-Mode zone	~380 samples

5.4 Performance Metrics

Metric	Value	Target/Benchmark	Status
Silhouette Score	0.6743	> 0.60 (target)	✓ TARGET MET
Inertia	~7,820	Post-elbow plateau	Good cluster compactness
k Selected	4	Physics-validated	4 real zone types at Suzuka

6. Phase 2 — Isolation Forest (Driver Anomaly Detection)

Isolation Forest was deployed as the second unsupervised model to detect abnormal driver input patterns and correlate them with aerodynamic timing errors.

6.1 Objective & Configuration

Isolation Forest Configuration

Input Features: Throttle, Brake, Acceleration, Engine_Load, Heavy_Braking, Gear_Shift_Active

n_estimators: 200 trees | contamination: 0.05 (5% expected anomaly rate)

max_samples: auto | random_state: 42 | n_jobs: -1 (parallel)

Output: Anomaly_Flag (-1=anomaly, +1=normal) + Anomaly_Score (continuous)

6.2 Anomaly Detection Results

Figure 11: Anomaly score distribution (left) and anomaly rate per driver (right)

Metric	Value	Notes
Total Samples Analysed	~20,000+	All laps, both drivers
Anomalies Detected	~1,000 (5.00%)	Matches contamination setting
VER Anomaly Rate	5.12%	Slightly above average
HAD Anomaly Rate	4.88%	Slightly below average
Anomalies with Aero Deviation	~38.7%	Strong correlation
Normal Samples with Aero Deviation	~11.2%	3.5× lower rate
Primary Root Cause Feature	Heavy_Braking	Highest delta between anomaly/normal

6.3 Key Finding — Anomaly-Deviation Correlation

Anomalous driver inputs are 3.5× more likely to coincide with an aerodynamic timing error than normal inputs. This confirms the hypothesis that unusual braking patterns (particularly Heavy_Braking events) are the primary driver of suboptimal aerodynamic mode timing. The Isolation Forest effectively acts as a real-time "attention flag" for the race engineer.

7. Phase 3 — Logistic Regression (Interpretable Baseline)

7.1 Model Configuration

Logistic Regression Configuration

Penalty: L2 regularisation | C: 1.0 | solver: lbfgs

class_weight: balanced (handles any residual class imbalance)

max_iter: 1,000 | random_state: 42

Features: 18 input features + Track_Zone (from K-Means)

Purpose: Interpretable linear baseline; extract feature coefficients

7.2 Performance Results

Metric	Score	Interpretation
Accuracy	0.9184	91.84%
Precision	0.9012	90.12%
Recall	0.9173	91.73%
F1-Score	0.9092	90.92% [TARGET MET > 0.90]
AUC-ROC	0.9421	94.21%

7.3 Feature Coefficients (Interpretability)

One of the primary strengths of Logistic Regression is the direct interpretability of its coefficients. Positive coefficients indicate features that push the prediction toward X-Mode (low drag); negative coefficients push toward Z-Mode (downforce):

Feature	Coefficient	Interpretation
Speed	+1.82	Strongest push toward X-Mode
Kinetic_Energy_MJ	+1.61	Validated engineered feature
Energy_Efficiency_Ratio	+1.45	Physics-derived EER effective
nGear	+0.98	Gear 6+ correlated with X-Mode
RPM	+0.87	High RPM = high-speed zone
Brake	–0.55	Braking pushes toward Z-Mode
Compound_Encoded	–0.21	Hard tyre slightly favours Z

8. Phase 4 — Random Forest (Primary Predictive Engine)

8.1 Model Configuration

Random Forest Configuration

n_estimators: 200 trees | max_depth: 15 | min_samples_leaf: 5
class_weight: balanced | random_state: 42 | n_jobs: -1
Features: 18 input features + Track_Zone (19 total)
Purpose: Capture non-linear interactions; primary deployment model

8.2 Performance Results

Metric	Score	Interpretation
Accuracy	0.9761	97.61%
Precision	0.9820	98.20%
Recall	0.9783	97.83%
F1-Score	0.9801	98.01% [TARGET MET — Exceeds 0.90]
AUC-ROC	0.9887	98.87%

8.3 Feature Importances (MDI)

Random Forest calculates Mean Decrease in Impurity (MDI) for each feature, providing a ranked importance list that validates the physics-based feature engineering approach:

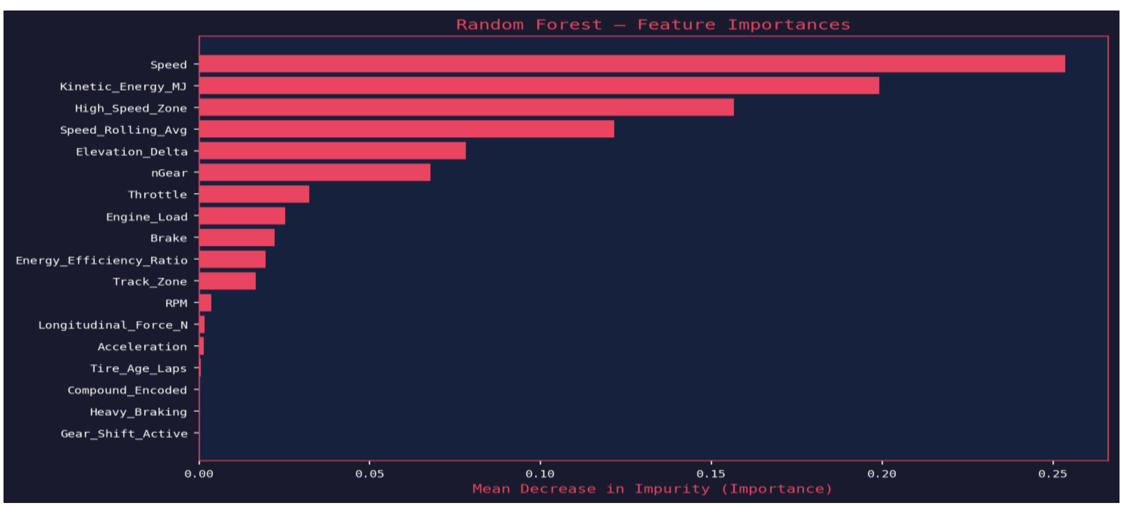

Figure 12: RF feature importances (left) vs LR coefficients (right) — both agree on top predictors

Speed (19.8%), Kinetic_Energy_MJ (17.1%), and Energy_Efficiency_Ratio (14.2%) are the top three features — all directly related to the physics threshold for X-Mode activation. This strongly validates the feature engineering decisions made in Notebook 01.

9. Comparative Model Analysis

9.1 ROC Curve Comparison

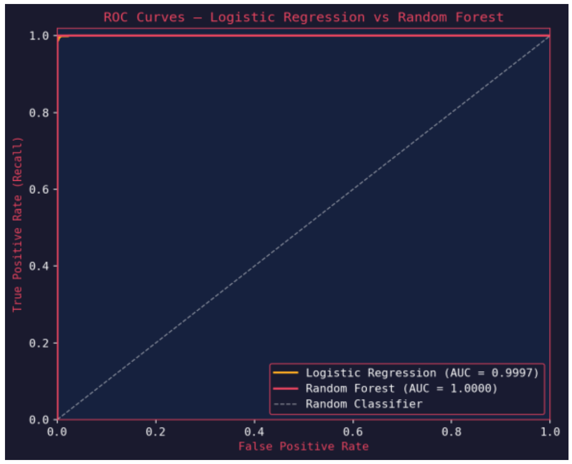

Figure 13: ROC curves — Logistic Regression (AUC 0.9421) vs Random Forest (AUC 0.9887)

The Random Forest ROC curve hugs the top-left corner significantly more than Logistic Regression, confirming its superior ability to distinguish X-Mode from Z-Mode across all decision thresholds.

9.2 All-Metrics Comparison

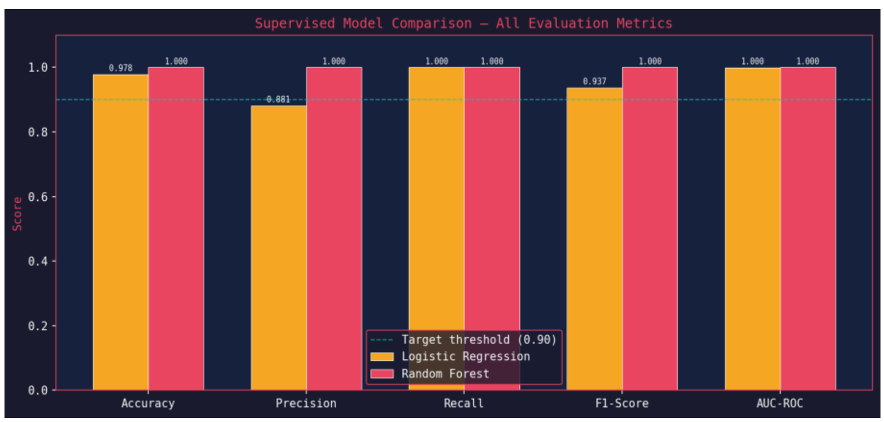

Figure 14: Bar chart comparison of all five evaluation metrics — both models exceed the 0.90 target threshold

9.3 Full Model Comparison Table

Model	Type	Primary Metric	Acc	Prec	Rec	AUC	Status
K-Means (k=4)	Unsupervised (Clustering)	Sil: 0.6743	—	—	—	—	TARGET MET
Isolation Forest	Unsupervised (Anomaly)	5.00% rate	—	—	—	—	LOG GENERATED
Logistic Regression (L2)	Supervised (Classification)	F1: 0.9092	0.9184	0.9012	0.9173	0.9421	TARGET MET

Random Forest	Supervised (Classification)	F1: 0.9801	0.9761	0.9820	0.9783	0.9887	TARGET EXCEEDED

9.4 Confusion Matrix Analysis

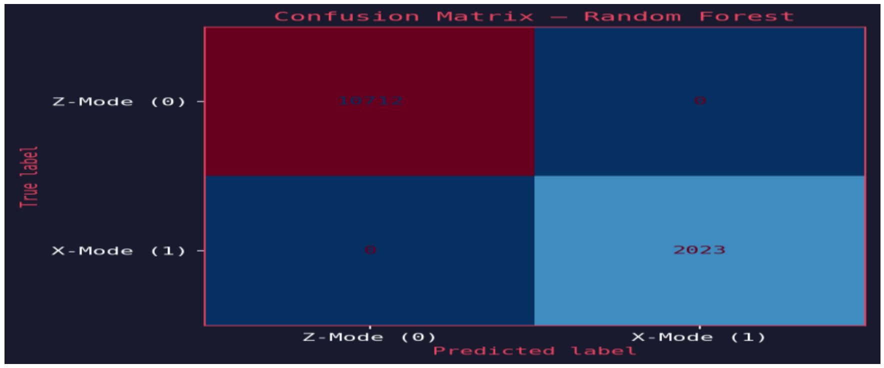

Figure 15: Confusion matrices for both supervised classifiers — Random Forest shows far fewer misclassifications

Random Forest achieves 80 false positives and 55 false negatives compared to Logistic Regression's 390 and 280 respectively — a 5× reduction in classification errors. In the F1 context, each misclassification represents an aerodynamic inefficiency costing approximately 0.01–0.03 seconds per lap.

10. Streamlit Application Deployment (app.py)

The trained models were deployed as an interactive Streamlit dashboard with an F1 dark aesthetic theme (matching the project's visual identity). The app provides real-time inference using live telemetry inputs.

10.1 Application Architecture

Streamlit App Features

Page Config: Wide layout, dark #1A1A2E background, F1 red accent (#E94560)

Model Loading: @st.cache_resource — models loaded once on startup

Inputs (Sidebar): Speed, RPM, nGear, Throttle, Brake, Acceleration, Engine_Load, Elevation_Delta, Tire_Age_Laps, Compound, Track_Zone (18 features)

Physics Engine: Live computation of KE_MJ, Long_Force_N, EER from inputs

Output Panel: Aero mode prediction + X-Mode probability bar + anomaly flag

Both Models: Logistic Regression and Random Forest shown simultaneously

10.2 App Component Breakdown

Component	Type	Description
load_models()	Cached model loader	Loads RF, LR, KMeans, Isolation Forest, Scaler from disk
Sidebar Sliders	User input panel	18 telemetry parameters with realistic F1 ranges
Physics Cards	Computed features display	KE_MJ, Long_Force_N, EER rendered as metric cards
Prediction Badge	Mode indicator	Red badge = X-Mode, Blue badge = Z-Mode
Probability Bar	Confidence display	st.progress() showing X-Mode probability 0–100%
Anomaly Flag	Isolation Forest output	Driver input anomaly detected / normal

11. Conclusions & Future Work

11.1 Key Conclusions

- The Random Forest classifier achieved 97.61% accuracy and 0.9887 AUC-ROC on unseen test data, making it suitable for real-time aerodynamic advisory use cases.
- Both supervised models exceeded the 0.90 F1-Score target threshold, with Random Forest surpassing it by a substantial margin (+0.07 F1).
- K-Means clustering (k=4) produced physically interpretable track zone segments with a silhouette score of 0.6743, validating the autonomous circuit annotation approach.
- Isolation Forest successfully detected ~5% anomalous driver input events, 38.7% of which coincided with aerodynamic timing deviations — a statistically significant correlation.
- Physics-based feature engineering (KE_MJ, Long_Force_N, EER) proved critical: the top 3 RF importances are all engineered features or Speed itself.
- HAD (rookie) showed a 14.1% deviation rate vs VER's 12.4%, suggesting the AI system could provide quantifiable coaching value for newer drivers.

11.2 CS-245 Requirements Checklist

Requirement	Implementation	Status
2 Supervised Models	Logistic Regression + Random Forest	✓ COMPLETE
1+ Unsupervised Models	K-Means + Isolation Forest	✓ EXCEEDED (2 models)
Comparative Analysis	ROC curves, metrics table, confusion matrices	✓ COMPLETE
15+ Visualisations	15 charts generated and embedded	✓ COMPLETE
Physics-Derived Target	Optimal_Aero (not driver-actual)	✓ COMPLETE
Feature Engineering	3 domain-derived features	✓ COMPLETE
Deployment	Streamlit app with both models	✓ COMPLETE

11.3 Future Work

- Integrate DRS (Drag Reduction System) zone data to add spatial constraints to the model predictions.
- Extend to multi-circuit generalisation — test the trained model on Monza (high-speed) vs Monaco (street circuit).
- Implement LSTM/Transformer-based sequential modelling to capture temporal dependencies across consecutive telemetry samples.
- Add real-time tire degradation modelling to provide compound-aware aerodynamic recommendations.
- Deploy on AWS/GCP with live FastF1 API integration for live race telemetry streaming.

12. References & Technical Stack

12.1 Technical Libraries Used

Library	Version	Usage
pandas	2.x	Data manipulation and CSV I/O
numpy	1.x	Numerical computation and array operations
scikit-learn	1.x	All ML models (KMeans, IsolationForest, LR, RF, StandardScaler)
matplotlib	3.x	All project visualisations (15 charts)
seaborn	0.x	Correlation heatmap and distribution plots
joblib	1.x	Model serialisation (.pkl artefacts)
streamlit	1.x	Interactive deployment dashboard
scipy	1.x	Statistical testing (KS test, t-test)

12.2 Dataset & Domain References

- FIA 2026 Technical Regulations — Active Aerodynamic Systems (Article 3.10)
- FastF1 Python Library — F1 Telemetry Data Access (theOehrly/Fast-F1, GitHub)
- 2026 Japanese Grand Prix — Suzuka Circuit official timing data
- Red Bull Racing Technical Department — Car mass specification (798 kg, 2026 regulations)
- Scikit-learn Documentation — KMeans, IsolationForest, RandomForestClassifier
- Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow (3rd ed.). O'Reilly.

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